

Technical Report

Term Deposit Customer Ranking

Purpose. Build a reproducible pre-call ranking workflow that helps a bank decide which already eligible customers should be called first about a term deposit.

40,000	7.24%	12
historical calls	subscription rate	valid pre-call predictors

Executive Summary

The project is a ranking problem rather than a conventional yes/no classification exercise. The bank has finite calling capacity, so the useful output is a pre-call score that orders already eligible customers from higher to lower historical propensity to subscribe. Accuracy is reported, but ranking quality, future-like robustness and call-depth lift drive the operational decision.

The target is strongly imbalanced: 2,896 of 40,000 historical calls ended in subscription (7.24%). The post-call variable **duration** is highly associated with the outcome (Pearson $r = 0.461169$) and substantially inflates retrospective model performance. It is therefore used only as leakage evidence and excluded from every deployable feature set, importance analysis, robustness test and lift calculation.

On the untouched 20% holdout, tuned HGB is the stronger conventional ranking model (PR-AUC / ROC-AUC = 0.272820 / 0.733930) than tuned LR (0.240265 / 0.709681). Under expanding-window ordered validation, however, LR is more robust (0.112626 / 0.566290) than HGB (0.096333 / 0.551600). LR also delivers the stronger period-local lift at tight call depth: 1.853857x at 5%, capturing 228 subscribers from 1,528 calls versus about 123 expected under random selection.

Decision

Use unweighted Logistic Regression with all 12 valid pre-call predictors as the primary controlled-pilot candidate. Retain HGB as a challenger. Do not treat raw scores as literal subscription probabilities and do not approve unrestricted production deployment from this retrospective dataset alone.

1. Decision Context and Prediction Moment

The deployable decision occurs **before a marketing call starts**. Valid predictors are age, job, marital, education, default, balance, housing, loan, contact, day, month and campaign. The output is a descending rank over an already eligible calling pool; consent, eligibility and contact-frequency rules remain outside the model.

A fixed 0.50 threshold is not the business objective. PR-AUC and ROC-AUC describe score ordering across thresholds; period-local lift tests whether the top-ranked fraction of later source-order blocks contains more subscribers than random calling at the same volume.

2. Data Audit and Exploratory Evidence

The canonical source has 40,000 rows, 13 original predictors and the target **y**. The labelled repository copy also carries a derived helper **y_binary**; each notebook drops any existing helper and recreates it from the original yes/no target before analysis.

Table 1: Dataset and target audit

Measure	Result
Rows	40,000
Subscribers	2,896
Non-subscribers	37,104
Subscription rate	7.24%
Majority-class accuracy	92.76%
Exact duplicate source rows	0
Conventional null values	0
Blank strings	0

Literal **unknown** categories are present in contact (12,765 records; 31.91%), education (1,531; 3.83%) and job (235; 0.59%). They are retained as explicit categories rather than silently imputed.

2.1 Direct pre-call associations

Numeric and categorical association are deliberately kept separate because their statistics are not commensurate. Pearson correlation is used for the four numeric pre-call variables; bias-corrected Cramer's V is used for categorical variables. Neither is treated as causal evidence or blended into a synthetic feature score.

Fig. 1: Pre-call numeric Pearson association with subscription.

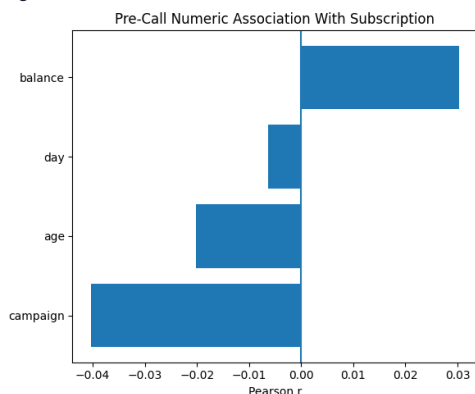
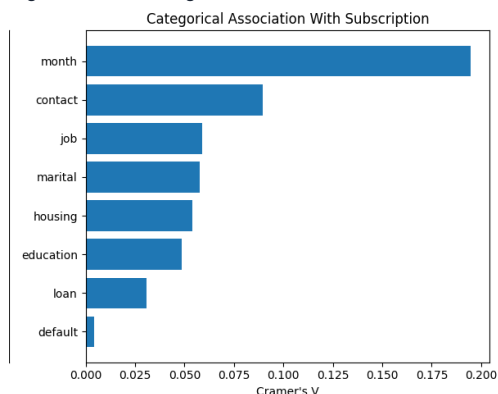


Fig. 2: Pre-call categorical Cramer's V association with subscription.

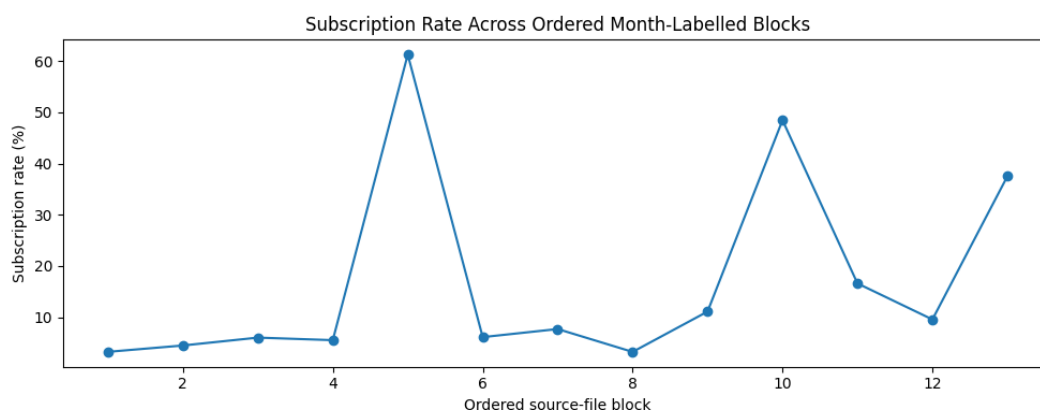


The numeric marginal associations are small: campaign -0.04035, balance +0.03023, age -0.02027 and day -0.00642. Categorical association is stronger for month (0.19453) and contact (0.08958). Weak marginal association does not imply a feature is useless after multivariable transformation or nonlinear interaction.

2.2 Ordered source blocks

The file has month labels but no verified year, campaign identifier or trustworthy timestamp. Consecutive changes in month label create 13 ordered source-file blocks. They are used as a future-like robustness approximation and must not be described as confirmed calendar months or years.

Fig. 3: Subscription Rate Across Ordered Month-Labelled Blocks.



3. Leakage Diagnostic: Call Duration

Post-call duration has Pearson $r = 0.461169$ with subscription, far larger than any legitimate pre-call numeric association. The same default model specification is therefore evaluated once with pre-call predictors and once with the identical settings plus duration. The resulting jump quantifies retrospective leakage rather than a deployable performance opportunity.

Table 2: Ranking impact of adding post-call duration

Stage	Model	Pre-call PR	+duration PR	PR change	Pre-call ROC	+duration ROC	ROC change
Default	LR	0.241676	0.525049	+0.283373	0.711156	0.932379	+0.221223
Default	HGB	0.268264	0.585190	+0.316926	0.734102	0.951528	+0.217425
Tuned	LR	0.240265	0.520693	+0.280427	0.709681	0.933188	+0.223507
Tuned	HGB	0.272820	0.549825	+0.277004	0.733930	0.943115	+0.209185

Leakage conclusion

Duration is excluded from deployable models, feature importance, feature reduction, imbalance experiments, robustness validation, calibration and lift. The report does not use a duration-based targeting recommendation.

4. Model Families, Validation and Hyperparameters

Two deliberately different model families are compared: Logistic Regression (LR) as the transparent linear reference and HistGradientBoosting (HGB) as the nonlinear challenger. The goal is a focused linear-versus-nonlinear test, not an exhaustive classifier tournament.

Using SEED=42, 32,000 rows form development data and 8,000 rows form an untouched final holdout. Inside development, LR tuning uses 24,000 training rows and 8,000 validation rows. HGB search uses a deterministic stratified 10,000-row subset of the tuning-training data. Hyperparameter selection uses this fixed train/validation design, **not K-fold CV**.

4.1 Hyperparameter definitions and search ranges

Table 3: Tuned hyperparameters, meaning and values searched

Model	Parameter	Meaning	Values searched
LR	C	Inverse regularisation strength	15 log-spaced values: 0.001 to 100
LR	penalty	Coefficient regularisation	L1, L2
LR	class_weight	Relative class weighting	None, balanced
HGB	max_iter	Boosting iterations	50, 100, 150, 200, 300, 400
HGB	max_depth	Maximum tree depth	3, 4, 5, 6, 8, None
HGB	learning_rate	Contribution per boosting step	0.01, 0.03, 0.05, 0.10, 0.20
HGB	max_leaf_nodes	Maximum leaves per tree	15, 31, 63, 127
HGB	l2_regularization	Leaf-value L2 penalty	0, 0.1, 0.5, 1, 2
HGB	min_samples_leaf	Minimum rows per leaf	10, 20, 30, 50
HGB	class_weight	Relative class weighting	None, balanced

LR exhaustively evaluates 60 candidates per feature condition. HGB has 28,800 possible combinations and reproducibly samples 100 per feature condition with random_state=42. Across LR pre-call, LR + duration, HGB pre-call and HGB + duration, exactly **320 tuned candidate configurations** are evaluated. PR-AUC is the primary selection objective and ROC-AUC the deterministic tie-breaker.

4.2 Baseline/default specifications

The critical baseline rule is simple: within each model family, **pre-call and +duration baselines use the same hyperparameters**. Only the feature set changes. This keeps the initial duration test a genuine feature-availability comparison.

Table 4: Baseline/default model settings - reused unchanged across feature conditions

Model	Baseline/default settings applied to both pre-call and +duration
LR	C=1.0; L2; solver=lbfgs; max_iter=1000; tol=1e-4; class_weight=None; seed=42
HGB	loss=log_loss; learning_rate=0.1; max_iter=100; leaves=31; depth=None; min_leaf=20; L2=0; max_bins=255; early_stopping=auto; class_weight=None; seed=42

4.3 Tuned specifications

Tuning is then run separately for each model x feature condition, producing four different selected winners. LR tuning fixes solver=liblinear, max_iter=3000 and seed=42 while searching C, penalty and class weight; HGB tuning uses early_stopping=True and seed=42 while searching the ranges in Table 3.

Table 5: Four selected tuned configurations

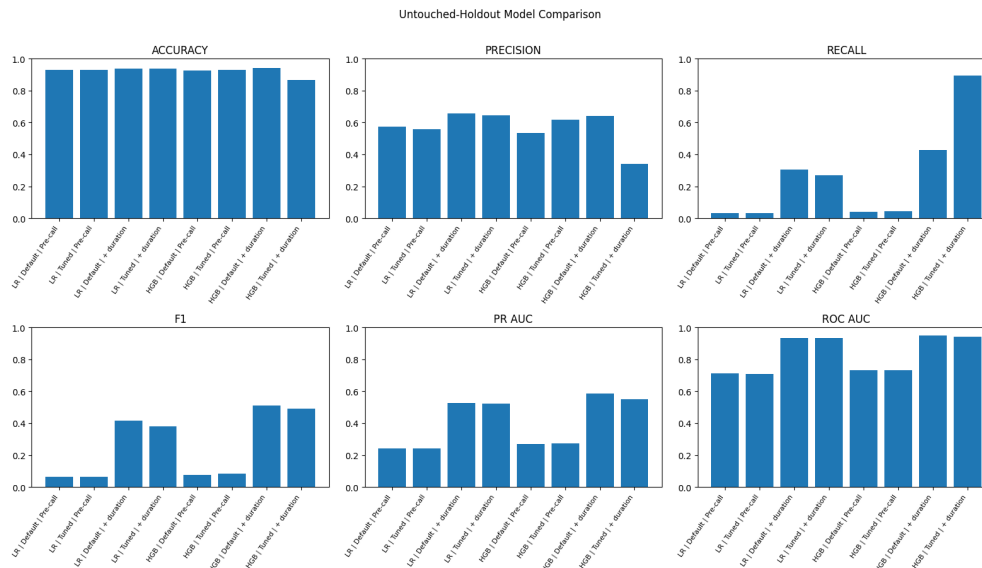
Configuration	Selected tuned settings
LR pre-call	L2; C=8.483428982440726; class_weight=None
LR + duration	L1; C=0.0610540229658533; class_weight=None
HGB pre-call	max_iter=200; depth=8; learning_rate=0.05; leaves=15; L2=0; min_leaf=20; class_weight=None; early_stopping=True
HGB + duration	max_iter=200; depth=3; learning_rate=0.05; leaves=127; L2=2; min_leaf=10; class_weight=balanced; early_stopping=True

5. Untouched Holdout Performance

Table 6: Default and tuned performance on the untouched 20% holdout

Stage	Model	Features	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
Default	LR	Pre-call	0.928250	0.575758	0.032815	0.062092	0.241676	0.711156
Tuned	LR	Pre-call	0.928125	0.558824	0.032815	0.061990	0.240265	0.709681
Default	HGB	Pre-call	0.928000	0.534884	0.039724	0.073955	0.268264	0.734102
Tuned	HGB	Pre-call	0.928875	0.619048	0.044905	0.083736	0.272820	0.733930
Default	LR	+duration	0.938125	0.655556	0.305699	0.416961	0.525049	0.932379
Tuned	LR	+duration	0.936500	0.646091	0.271157	0.381995	0.520693	0.933188
Default	HGB	+duration	0.941125	0.639896	0.426598	0.511917	0.585190	0.951528
Tuned	HGB	+duration	0.866000	0.338572	0.892919	0.490978	0.549825	0.943115

Fig. 4: Six-metric untouched-holdout comparison for default/tuned LR and HGB, with and without post-call duration.



Tuning changes valid pre-call ranking only modestly. LR PR-AUC moves from 0.241676 to 0.240265 and HGB from 0.268264 to 0.272820. The much larger gains occur only when post-call duration is introduced. This is why the final model decision cannot be based on the highest retrospective holdout metric alone.

6. Feature Evidence and Reduction

Direct association, LR coefficient magnitude and HGB permutation importance answer different questions and are kept separate. Feature reduction is ordered **only** by HGB held-out permutation importance measured as mean PR-AUC decrease after permuting each original predictor.

Fig. 5: Logistic Regression grouped coefficient magnitude by original feature.

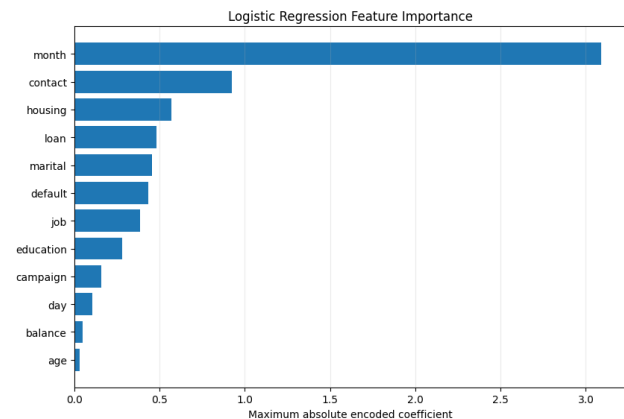
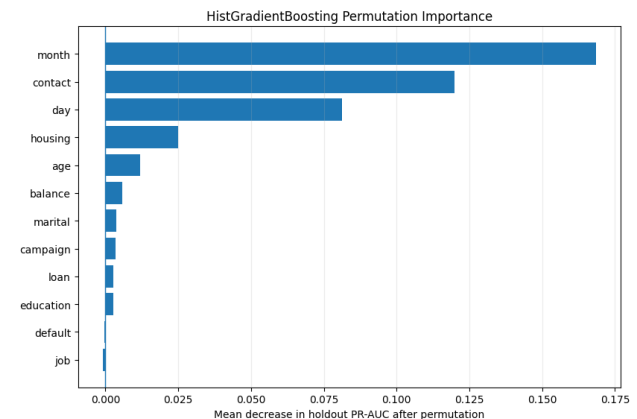


Fig. 6: HistGradientBoosting held-out permutation importance by original feature.



The LR model is dominated by month at the encoded-coefficient scale, while HGB permutation importance ranks month, contact and day most strongly. Negative values for default and job indicate negligible or slightly adverse estimated PR-AUC change under the permutation experiment; they are not proof that those variables are intrinsically harmful in every future campaign.

Table 7: HGB permutation ranking used for feature reduction

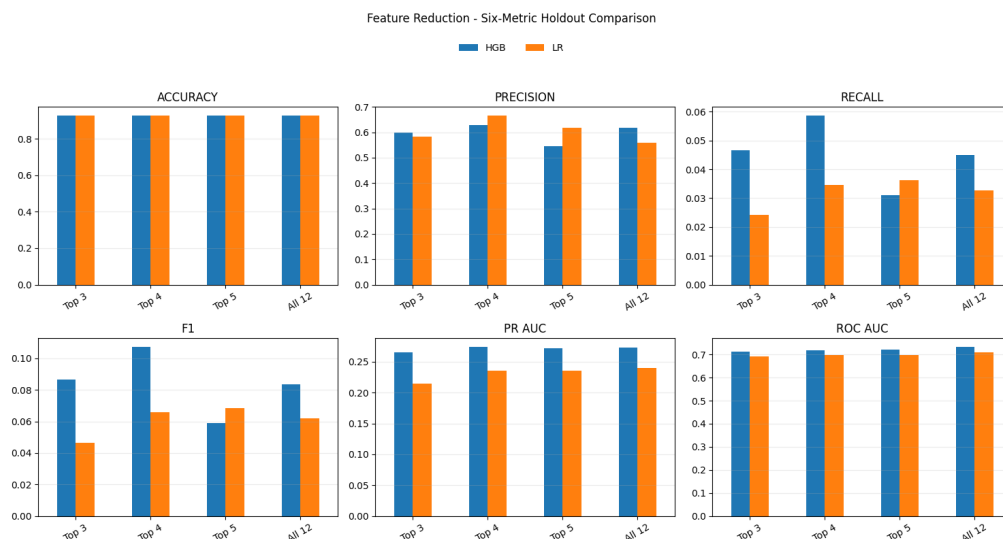
Rank	Feature	PR-AUC decrease
1	month	0.168555
2	contact	0.119955
3	day	0.081196
4	housing	0.025086
5	age	0.011867
6	balance	0.005755
7	marital	0.003635
8	campaign	0.003570
9	loan	0.002746
10	education	0.002696
11	default	-0.000307
12	job	-0.000900

Top-3 = month/contact/day; Top-4 adds housing; Top-5 adds age. No reduced subset is returned, so the experiment isolates predictor removal rather than a second hyperparameter search.

Table 8: Feature-reduction holdout comparison

Set	Model	Accuracy	F1	PR-AUC	ROC-AUC
Top 3	LR	0.928125	0.046434	0.214884	0.691444
Top 3	HGB	0.928750	0.086538	0.265716	0.713004
Top 4	LR	0.928875	0.065681	0.235485	0.698631
Top 4	HGB	0.929375	0.107425	0.274531	0.719196
Top 5	LR	0.928625	0.068515	0.235275	0.699099
Top 5	HGB	0.928000	0.058824	0.272162	0.720062
All 12	LR	0.928125	0.061990	0.240265	0.709681
All 12	HGB	0.928875	0.083736	0.272820	0.733930

Fig. 7: Six-metric feature-reduction comparison using the fixed tuned pre-call structures.



HGB Top-4 has slightly higher PR-AUC than All-12 (0.274531 vs 0.272820), but All-12 has materially stronger ROC-AUC (0.733930 vs 0.719196). A small exploratory holdout difference is insufficient evidence to declare the reduced subset universally superior; the pilot specification retains all 12 valid pre-call predictors.

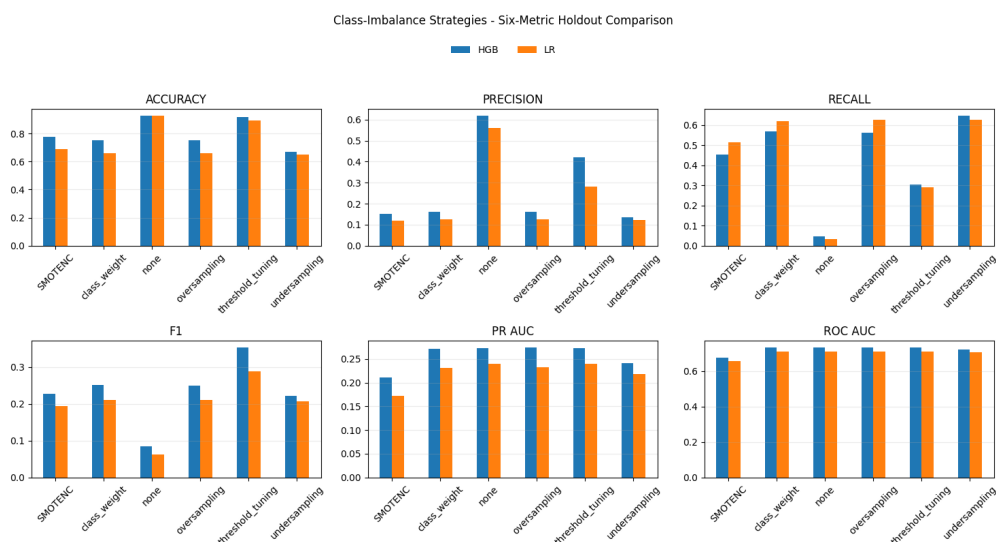
7. Class Imbalance and Threshold Behaviour

Six strategies are evaluated on the tuned pre-call model structures: no adjustment, balanced class weighting, random oversampling, random undersampling, SMOTENC and threshold tuning. Thresholds are selected from five-fold out-of-fold development predictions and then applied once to the untouched holdout.

Table 9: Class-imbalance results

Model	Strategy	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
LR	none	0.928125	0.558824	0.032815	0.061990	0.240265	0.709681
LR	class weight	0.662875	0.126587	0.620035	0.210249	0.231243	0.710575
LR	oversampling	0.661000	0.126967	0.626943	0.211169	0.232024	0.710790
LR	undersampling	0.651875	0.123549	0.625216	0.206327	0.218951	0.708137
LR	SMOTENC	0.689875	0.119295	0.514680	0.193695	0.172603	0.656204
LR	threshold 0.141	0.895125	0.282609	0.291883	0.287171	0.240265	0.709681
HGB	none	0.928875	0.619048	0.044905	0.083736	0.272820	0.733930
HGB	class weight	0.754750	0.161527	0.569948	0.251716	0.271896	0.732946
HGB	oversampling	0.753625	0.159491	0.563040	0.248570	0.274559	0.733041
HGB	undersampling	0.671750	0.134072	0.647668	0.222156	0.241083	0.723593
HGB	SMOTENC	0.775250	0.150716	0.454231	0.226334	0.210334	0.674565
HGB	threshold 0.156	0.919375	0.421053	0.303972	0.353059	0.272820	0.733930

Fig. 8: Six-metric class-imbalance comparison for LR and HGB.



Weighting and resampling primarily change precision/recall trade-offs rather than producing a robust ranking advantage. HGB oversampling increases PR-AUC by only 0.001739 while sharply reducing accuracy and precision. SMOTENC degrades ranking for both families. Threshold tuning produces the strongest hard-classification F1 (LR 0.287171; HGB 0.353059) but leaves PR-AUC and ROC-AUC unchanged because score ordering is unchanged.

8. Exploratory Population and Robustness

A descriptive higher-conversion population is defined as **contact=cellular AND (job=retired OR age>=71 OR balance>=Q3)**. With Q3 balance = 1,319, it contains 6,774 records, 778 subscribers and an 11.485% historical conversion rate versus 7.24% overall. This remains contextual evidence, not a replacement targeting rule.

When the Q3 threshold is learned within each validation fold and default full-population models are used without subgroup retuning, validated population PR-AUC is 0.286349 for LR and 0.303476 for HGB. The result does not establish causal benefit or justify a separate demographic model.

8.1 Shuffled, pseudo-profile and ordered validation

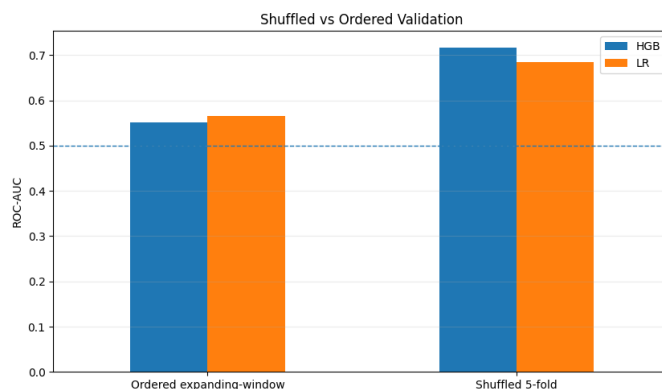
Table 10: Robustness comparison

Validation	LR PR-AUC	LR ROC-AUC	HGB PR-AUC	HGB ROC-AUC
Shuffled 5-fold	0.215433	0.684767	0.245333	0.717461
Pseudo-profile grouped sensitivity	0.215690	0.685003	0.248093	0.716737
Ordered expanding-window	0.112626	0.566290	0.096333	0.551600

Pseudo-groups are exact age + job + marital + education + balance combinations. Similarity to shuffled validation is reassuring as a sensitivity check, but no true customer identifier exists, so this cannot prove customer-level independence.

Ordered validation trains on eligible earlier source blocks and tests on a later block. A test block is included only when at least 5,000 earlier rows exist and the block contains at least 100 positive and 100 negative outcomes. Eight blocks qualify, covering 30,526 test rows and 2,457 subscribers.

Fig. 9: Shuffled vs Ordered Validation.



Central robustness result

HGB is stronger under conventional shuffled historical validation; LR is stronger under the more demanding ordered expanding-window test. The large degradation for both models is the principal reason the analysis supports controlled pilot planning rather than unconditional production deployment.

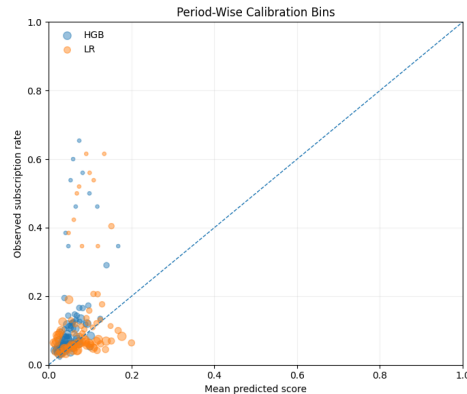
9. Calibration Diagnostics and Capacity-Based Ranking

Forward predictions are diagnosed with period-local quantile bins, Brier score and expected calibration error (ECE). No sigmoid, isotonic or CalibratedClassifierCV model is fitted. Raw scores are therefore interpreted as ranking signals rather than trustworthy individual subscription probabilities.

Table 11: Ordered calibration summary

Model	Rows	Base rate	Mean score	Brier	ECE	ROC-AUC	PR-AUC
LR	30,526	0.080489	0.075410	0.074776	0.044442	0.566290	0.112626
HGB	30,526	0.080489	0.048578	0.074181	0.035101	0.551600	0.096333

Fig. 10: Period-wise calibration bins for forward predictions; the diagonal denotes perfect calibration.

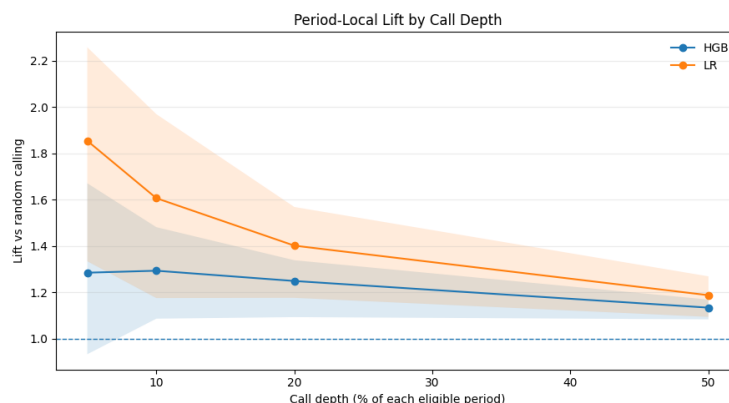


Period-local lift ranks customers separately within each eligible ordered block, takes the top call-depth fraction and compares captured subscribers with random calling at the same volume. The historical call depths below are sensitivity points, not prescribed production quotas.

Table 12: Period-local call-depth lift

Model	Depth	Called	Captured	Random expected	Lift	95% CI
LR	5%	1,528	228	122.99	1.853857	1.333945-2.257328
LR	10%	3,055	395	245.89	1.606389	1.175284-1.968856
LR	20%	6,108	689	491.63	1.401474	1.176270-1.568156
LR	50%	15,264	1,459	1,228.58	1.187549	1.094145-1.268835
HGB	5%	1,528	158	122.99	1.284690	0.932990-1.670341
HGB	10%	3,055	318	245.89	1.293245	1.086337-1.481000
HGB	20%	6,108	614	491.63	1.248918	1.093008-1.338290
HGB	50%	15,264	1,393	1,228.58	1.133829	1.083534-1.168554

Fig. 11: Period-Local Lift by Call Depth.



At 5% depth, LR captures 228 subscribers from 1,528 calls versus approximately 122.99 expected randomly. Its 95% interval is entirely above 1.0, while HGB at 5% has a wide interval crossing 1.0 (0.932990-1.670341). The bank should choose actual call depth from staffing capacity, contact cost and net contribution rather than adopting a fixed historical percentage.

10. Final Model Selection and Controlled Pilot

The primary pilot candidate is **unweighted Logistic Regression using all 12 valid pre-call predictors**; tuned unweighted HGB remains the challenger. LR is not chosen because it wins every retrospective metric - it does not. HGB wins conventional holdout and shuffled ranking, while LR is stronger under ordered validation and materially stronger at tight historical call depth.

Table 13: Decision evidence

Evidence	LR	HGB	Interpretation
Tuned holdout PR-AUC	0.240265	0.272820	HGB stronger conventional holdout ranking
Shuffled 5-fold ROC-AUC	0.684767	0.717461	HGB stronger when history is shuffled
Ordered weighted ROC-AUC	0.566290	0.551600	LR stronger source-order robustness
Ordered weighted PR-AUC	0.112626	0.096333	LR stronger ordered ranking
5% period-local lift	1.853857	1.284690	LR stronger at tight call capacity
Operational transparency	Higher	Lower	LR easier to inspect, monitor and challenge

A prospective pilot should allocate equal calling capacity to the LR-ranked arm and a business-as-usual or random comparator; an HGB challenger arm may be added if capacity permits. The primary outcome should be subscriptions per 1,000 assigned calls. Secondary controls should include contactability, complaints, opt-outs, repeat contacts, customer mix and agent time where available.

Decision gate

APPROVE for controlled pilot planning. DO NOT APPROVE for unrestricted production deployment. Prospective evidence is still required for incremental value, stability, calibration and economics.

11. Limitations and Required Next Evidence

Temporal uncertainty. Ordered blocks preserve source order but are not verified campaign timestamps or calendar years. **Customer identity.** No true customer ID exists; pseudo-profile grouping is only a sensitivity check. **Calibration.** No prospective recalibration model has been fitted. **Causality.** Historical association and lift do not estimate the treatment effect of calling. **Economics.** The dataset does not contain real call cost or net contribution per new deposit, so the project does not invent ROI.

A defensible economic pilot calculation should use measured incremental subscriptions and actual bank economics: **incremental value = incremental subscriptions x net contribution per subscription - incremental operating cost**. If customer-level expected value is required, prospective score recalibration should precede probability-based optimisation.

12. Reproducibility and Evidence Traceability

The repository is intentionally flat. All five notebooks read **term-deposit-marketing-2020-labelled.csv** from repository root, recreate the helper target, use SEED=42, contain explanatory Markdown between substantive code stages, and do not require cached CSV/JSON/NPY outputs, generated results/figures folders or HTML exports.

Table 14: Notebook-to-report traceability

Notebook	Role	Report evidence
01_data_eda.ipynb	Data audit, associations, ordered blocks	Tables 1-2; Figs. 1-3; duration timing evidence
02_baseline_tuning.ipynb	Baselines, tuning, holdout	Tables 2-6; Fig. 4; exact default/tuned parameters
03_features_imbalance.ipynb	Importance, reduction, imbalance	Tables 7-9; Figs. 5-8
04_segments_robustness.ipynb	Population and robustness	Table 10; Fig. 9
05_final_validation_reporting.ipynb	Calibration and lift	Tables 11-12; Figs. 10-11

Cross-notebook reconciliation remains exact: Notebook 2 tuned pre-call metrics match Notebook 3 All-12 metrics; Notebook 4 ordered metrics match Notebook 5 ordered/calibration summaries; the forward sample is consistently eight eligible blocks, 30,526 rows and 2,457 positives. The prior cross-notebook audit recorded 99 passes and zero failures.

Pinned environment: numpy 2.3.5, pandas 2.2.3, scipy 1.17.0, matplotlib 3.10.8, scikit-learn 1.8.0, imbalanced-learn 0.14.1, nbconvert 7.17.1 and ipykernel 7.3.0. Reproduce in notebook order 01 -> 02 -> 03 -> 04 -> 05. Notebook 2 performs the search; downstream notebooks use the selected pre-call settings rather than silently retuning.

12.1 Metric interpretation

Metric	What it answers in this project
Accuracy	How often the 0.50 hard-class prediction is correct; weak evidence alone because 92.76% of rows are non-subscribers.
F1	Balance of precision and recall at a chosen hard threshold; useful for threshold diagnostics, not the ranking policy.
PR-AUC	How well subscribers are concentrated toward higher scores across thresholds; primary tuning metric because positives are rare.
ROC-AUC	How often a subscriber is ranked above a non-subscriber across thresholds; useful complementary ranking measure.
Brier / ECE	Diagnostics of score reliability/calibration; lower is better, but no prospective recalibration model is fitted here.
Lift	Subscriber enrichment in the top-ranked call fraction relative to random calling at the same volume; directly linked to capacity use.

Evidence standard

All analytical tables and figures in this report are traceable to the final notebook calculations or notebook-generated visual outputs. No confusion matrices, post-call deployment features, sigmoid/isotonic calibration claims, report-only predictive analyses or cached historical result files are used.

Final analytical rerun and report consolidation: 26 August 2026